EXPLANATORY DATA ANALYSIS · TITANIC DATASET

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Who survived the TITANIC?

Survival was not random. Women and passengers in the upper classes were far more likely to survive. while most of those who died were men travelling in third class.

38%

Survived overall

342 of 891 passengers

4x

Women survived at **almost 4 times** the rate of men

2.6x

1st class survived at **2.6 times** the rate of 3rd class

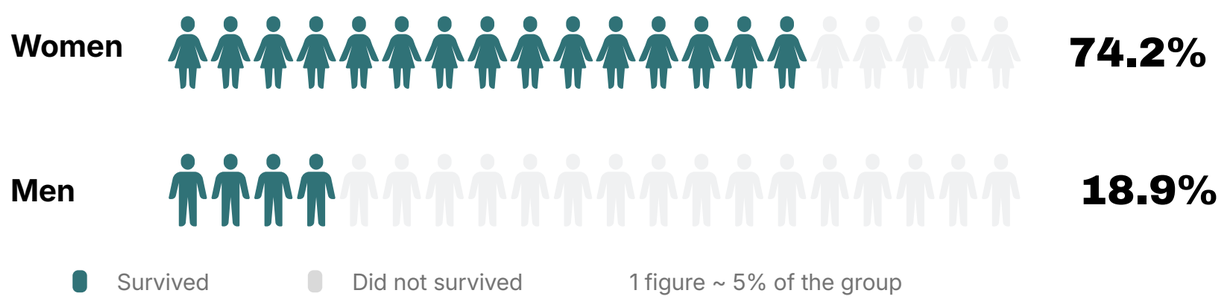
28

median age of both survivors and non-survivors

H2

Sex: women First

Out of 20 passengers, how many survived?

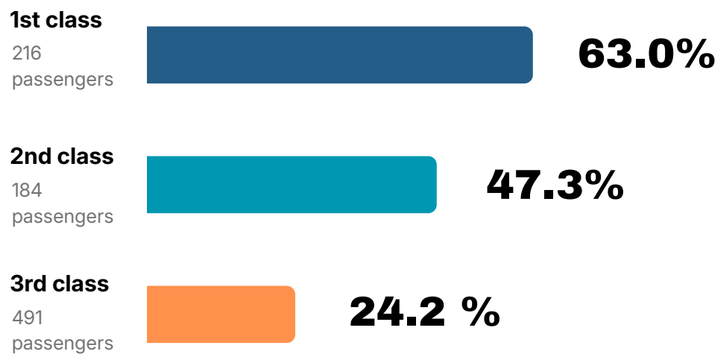


The biggest difference of all. Women were put first in the lifeboats (“women and children first”). The statistical test confirms this is real pattern, not chance.

H1

Class: the higher, the safer

Survival rate by passenger class

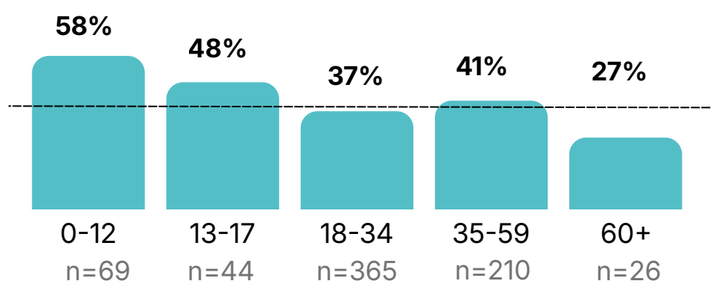


More than half of the passengers travelled in 3rd class, yet they had the lowest survival rate. Their cabins were mostly on the lowers decks, far from the lifeboats

H3

Age: it only mattered for children

Survival by age group ~ dashed line = 38% average



What age tell us about survival

For most people, age did not matter. The typical age was 28 for both survivors and those who died, and the statistical test did not find a clear difference between the two groups.

The exception: young children. More than half of children aged 0-12 survived (58%), well above the 38% average .
They were a small group, so they do not change the overall picture, but they were clearly given priority.

DID THE STATISTICS CONFIRM WHAT THE CHARTS SHOW?

Class - Yes

Real differences between classes

Sex - Yes

The strongest pattern of all

Age - Not overall

Only young children stand out

TYPICAL SURVIVOR

A woman travelling in 1st and 2nd class

or a young child of any class

TYPICAL VICTIM

An adult man travelling in 3rd class

only 18.9% of men and 24.2% of 3rd-class passengers survived

Conclusion: Survival followed a clear order: being a woman mattered most, then travelling in a higher class, then being a young child. **Keep in mind:** The data shows patterns not causes (for example, class is also linked to where cabins were on the ship), 177 passengers (19.9%) had no recorded age, and the data covers 891 of about 2,200 people aboard. **Method:** Survival rates compared across groups; tests used: chi-square (class, sex) and Mann - Whitney U (age), with a difference counted as real when $p<0.05$